1. Let P denote the plane spanned by $(1, 4, -1, -1)$ and $(1, 4, 1, 4)$ in $\mathbf{R}^4$

   1.1. Find a basis for $P^\perp$

   let $A = \begin{bmatrix} 1 & 4 & -1 & -1 \\ 1 & 4 & 1 & 4 \end{bmatrix} \Rightarrow RRE(A) = \begin{bmatrix} 1 & 4 & 0 & 1.5 \\ 0 & 0 & 1 & 2.5 \end{bmatrix}$

   $\therefore$ The basis for $P^\perp$ is $\left\{ \begin{pmatrix} -4 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -1.5 \\ 0 \\ -2.5 \\ 1 \end{pmatrix} \right\}$

   1.2. Find an implicit description of $P^\perp$

   The implicit description of $P^\perp$ is $nul(A)$

2. Consider the plane $2x - 3y + z = 0$ in $\mathbf{R}^3$

   2.1. Find a basis for its orthocomplement.

   The basis for $P^\perp$ is $\left\{ \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} \right\}$

   2.2. Find an implicit description of its orthocomplement.

   $\left[\begin{array}{c|c} 2 & b_1 \\ -3 & b_2 \\ 1 & b_3 \end{array}\right] \rightarrow \left[\begin{array}{c|c} 1 & 0.5b_1 \\ 0 & 1.5b_1 + b_2 \\ 0 & -0.5b_1 + b_3 \end{array}\right] \Rightarrow \begin{array}{l} 1.5b_1 + b_2 = 0 \\ -0.5b_1 + b_3 = 0 \end{array}$

   The implicit description of $P^\perp$ is $nul\left(\begin{bmatrix} 1.5 & 1 & 0 \\ -0.5 & 0 & 1 \end{bmatrix}\right)$

3. Let S denote the nullspace of the matrix below, find the bases for S and $S^\perp$.

   $S = nul(A)$, $S^\perp = row(A)$

   $A = \begin{bmatrix} 4 & 2 & 4 & 1 \\ -2 & 4 & 2 & 1 \\ 1 & 2 & 2 & 1 \\ 2 & 2 & 2 & 2 \end{bmatrix} \Rightarrow RRE(A) = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1.5 \\ 0 & 0 & 1 & -1.5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

   $\therefore$ The basic for S is $\left\{ \begin{pmatrix} -1 \\ -1.5 \\ 1.5 \\ 1 \end{pmatrix} \right\}$, The basic for $S^\perp$ is $\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1.5 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ -1.5 \end{pmatrix} \right\}$

4. Find a basis for the orthocomplement of the line spanned by $(1, 1, 1, 1, 1) \subset \mathbf{R}^5$.

L

$$L^{\perp} = \text{perp}\left\{\begin{pmatrix}1\\1\\1\\1\\1\end{pmatrix}\right\} = \text{nul}\begin{bmatrix}1 & 1 & 1 & 1 & 1\end{bmatrix}$$

$$\therefore \text{ The basis for } L^{\perp} \text{ is } \left\{\begin{pmatrix}-1\\1\\0\\0\\0\end{pmatrix}, \begin{pmatrix}-1\\0\\1\\0\\0\end{pmatrix}, \begin{pmatrix}-1\\0\\0\\1\\0\end{pmatrix}, \begin{pmatrix}-1\\0\\0\\0\\1\end{pmatrix}\right\}$$

5. Find bases for each of the four subspaces associated with the given matrix below.

$$A = \begin{bmatrix}2 & 4 & 4 & -2\\1 & 2 & 4 & -3\\1 & 2 & 3 & -2\end{bmatrix} \Rightarrow \text{RRE}(A) = \begin{bmatrix}1 & 2 & 0 & 1\\0 & 0 & 1 & -1\\0 & 0 & 0 & 0\end{bmatrix}$$

$$\text{RRE}(A^T) = \begin{bmatrix}1 & 0 & 0.25\\0 & 1 & 0.5\\0 & 0 & 0\\0 & 0 & 0\end{bmatrix}$$

$\therefore$ The basis for $\text{col}(A) = \left\{\begin{pmatrix}2\\1\\1\end{pmatrix}, \begin{pmatrix}4\\4\\3\end{pmatrix}\right\}$, The basis for $\text{row}(A) = \left\{\begin{pmatrix}1\\2\\0\\1\end{pmatrix}, \begin{pmatrix}0\\0\\1\\-1\end{pmatrix}\right\}$

The basis for $\text{nul}(A) = \left\{\begin{pmatrix}-2\\1\\0\\0\end{pmatrix}, \begin{pmatrix}-1\\0\\1\\1\end{pmatrix}\right\}$, The basis for $\text{nul}(A^T) = \left\{\begin{pmatrix}-0.25\\-0.5\\1\end{pmatrix}\right\}$

6. Given the matrix $\mathbf{A} = \begin{bmatrix}1 & 5 & -3\\1 & -3 & 5\end{bmatrix} \Rightarrow \text{RRE}(A) = \begin{bmatrix}1 & 0 & 2\\0 & 1 & -1\end{bmatrix}$

6.1. Find sets of vectors R and C that span $\text{row}(\mathbf{A})$ and $\text{col}(\mathbf{A})$, respectively.

$$C = \left\{\begin{pmatrix}1\\1\end{pmatrix}, \begin{pmatrix}5\\-3\end{pmatrix}\right\}, \quad R = \left\{\begin{pmatrix}1\\0\\2\end{pmatrix}, \begin{pmatrix}0\\1\\-1\end{pmatrix}\right\}$$

6.2. Find a set V such that $\text{row}(\mathbf{A}) = V^{\perp}$.

$$V = \text{nul}(A) \Rightarrow V = \left\{\begin{pmatrix}-\frac{1}{2}\\1\\1\end{pmatrix}\right\}$$

6.3. What is the dimension of $\text{col}(\mathbf{A})^{\perp}$?

$$= \dim(\text{nul}(A^T)) = n - r = 2 - 2 = 0$$